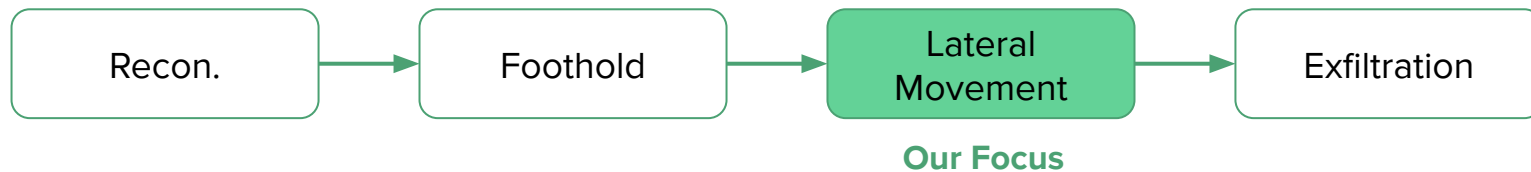


Real-Time Detection of Lateral Movement in Cloud VPC Networks via Graph-Based Analysis

Lateral Movement Goes Undetected



- Internal traffic is trusted
- Single log is not enough
- Volume drowns out the signals

Graph-Based Multi-Source Detection

Nodes

Computers and user identities

Edges

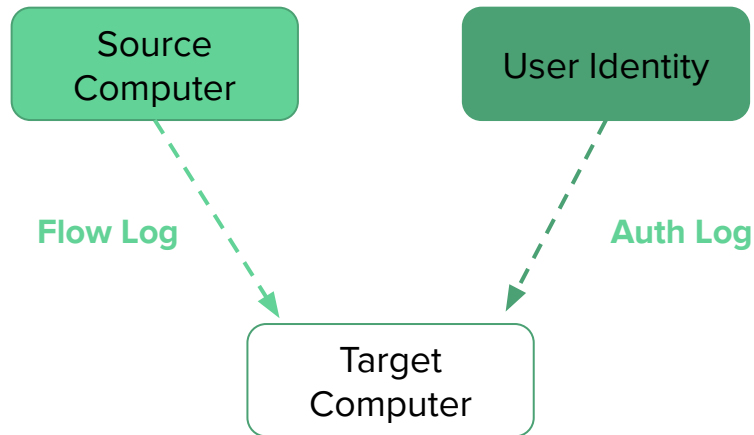
Network connections from flow logs and authentication events

Features

Degree, centrality, fan-out, timing / length of connection, etc.

2 sources of Log Input:

- Flow data alone misses who logged in
- Authentication data misses recon.



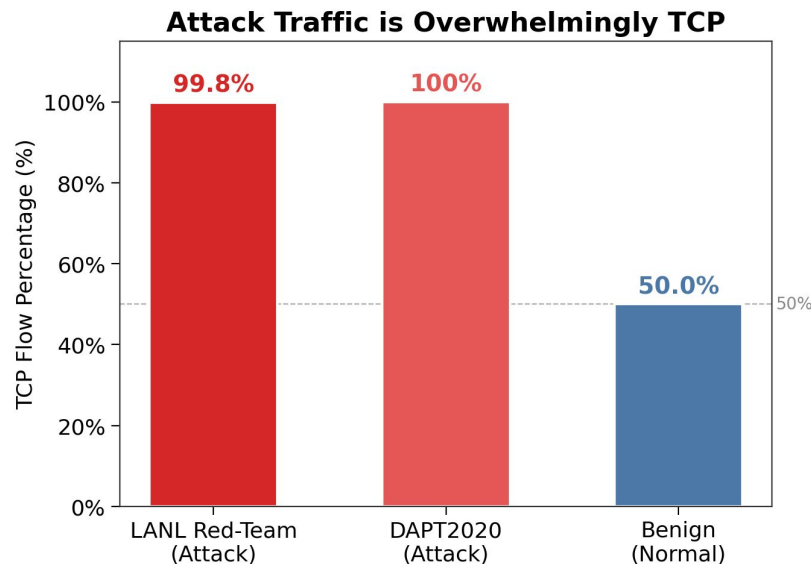
Datasets - Real-World and Simulated Attack Data

	LANL-Dataset-2015	DAPT2020
Type	Real enterprise network	Simulated APT Testbed
Capture Period	58 days Continuous	5 days
Network Size	1.6B+ events total	20.665 attack flows
Attack Ground Truth	749 labeled red-team events	Per-flow labels for 4 kill-chain
Log Sources	Flows & Auth & DNS & Processes	PCAP Derived Flows & System logs

Graph Scoring

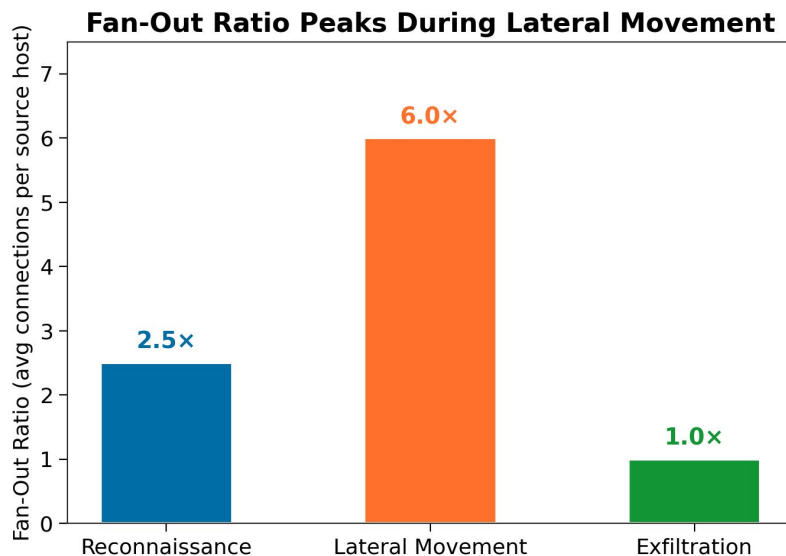
- Score nodes/edges based on feature deviation from baseline
 - Flag subgraphs above threshold
- Multiple different scoring approaches
 - Flow only
 - Auth only
 - Combination

Analysis of Data - Traffic Protocols



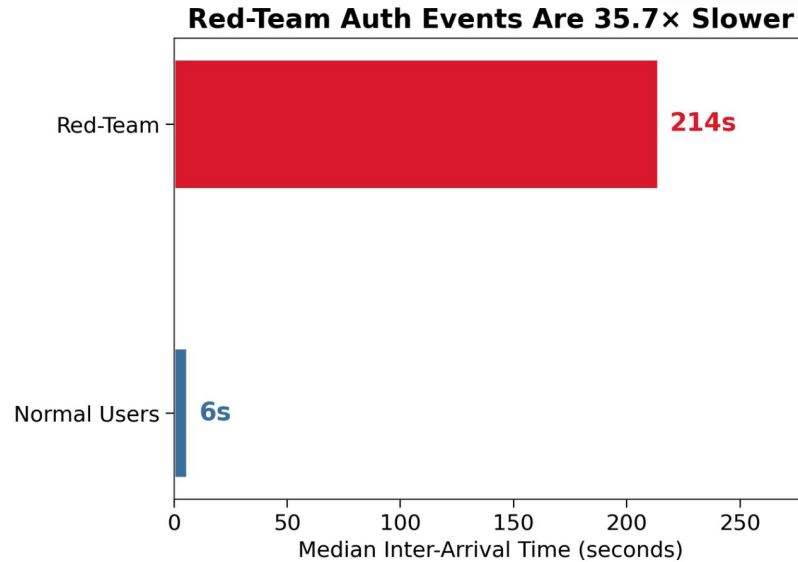
- 99.82% of LANL red-team flows and 100% of DAPT2020 attack flows use TCP
 - HTTP/3 uses UDP instead of TCP (since 2021)!
 - Interesting to see if the data has changed since then.

Analysis of Data - Fan-out Ratio



- Unlike benign traffic which is ~50% TCP, Fan-out ratio increases from 2.50 (reconnaissance) to 6.0 (lateral movement), then drops to 1.0 (exfiltration)

Analysis of Data - Inter-Arrival Time



- Red-team authentication events have median inter-arrival time of 214 seconds vs 6 seconds for normal users (35.7x difference)

Preliminary Results

Method	Recall	FPR	F1	AUC
<i>Graph-based (pair space)</i>				
combined	0.662	0.020	0.039	0.954
auth_only	0.747	0.030	0.036	0.951
flow_only	0.013	0.030	0.002	0.579
<i>Unsupervised tabular baselines (edge space)</i>				
Isolation Forest	0.741	0.050	0.024	0.940
Elliptic Envelope	0.000	0.050	0.000	0.774
One-Class SVM	0.492	0.098	0.008	0.706
PCA Reconstruction	0.033	0.050	0.001	0.667
Local Outlier Factor	0.207	0.045	0.008	0.493

Future Plans

- Ingest all 1.6B+ events within our pipeline
- Tweak which features are more useful / less useful.
- Compile statistics

Graph-based Analysis Is Useful!

- Ingest data into graphs with nodes, features
- Score each path to determine 'attack' subgraphs
- Detect threats faster, with no dependence on ML models.
 - Easily adjustable later on, depending on which features become more of an issue!